

Selected Database: PostgreSQL Database - Amazon RDS

Database type: Relational SQL

Deployment: AWS

Justification: We will use PostgreSQL because our application contains structured, relational data such as users, teams, projects, EOIs and allocations. PostgreSQL supports primary and foreign keys, relationships and constraints, which aligns directly with our database schema and ERD. It is also open-source, widely supported, scalable, and can be deployed through AWS.

Database Schema (v1):

Table	Main fields
Users	user_id PK, full_name, email, password_hash, role, created_at
Teams	team_id PK, team_name, created_at
Team Members	team_member_id PK, user_id FK, team_id FK
Projects	project_id PK, sponsor_id FK, title, description, required_skills, status, created_at
StudentEOIs	student_eoi_id PK, team_id FK, project_id FK, majors, skillset, status, submitted_at
MentorEOIs	mentor_eoi_id PK, user_id FK, experience, interests, availability, status, submitted_at
SponsorEOIs	sponsor_eoi_id PK, user_id FK, project_title, description, scope, requirements, company_details, status, submitted_at
ProjectAllocations	allocation_id PK, project_id FK, team_id FK, mentor_id FK, allocated_at
Notifications	notification_id PK, user_id FK, notification_type, message, is_read, created_at
TeamCharters	charter_id PK, team_id FK, file_path, uploaded_at
Resources	resource_id PK, title, description, resource_type, link, created_at
PastProjects	past_project_id PK, title, description, outcome, sponsor_name, mentor_name, year